Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA101-1 **Max Marks: 100**

Course Name: FOUNDATIONAL MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Reformulate statements from common language to formal logic using the rules of propositional and predicate calculus and assess the validity of arguments.

CO2: formulate and interpret statements presented and determine their validity by applying the rules and methods of propositional logic.

CO3: construct elementary proofs using ordinary and strong induction in the context of studying the properties of recursion.

CO4: apply basic counting principles including the pigeonhole principle and rules for counting permutations and combinations.

CO5: apply the logical structure of proofs and work symbolically with connectives and quantifiers to produce logically valid, correct and clear arguments

Section A **3*8=24**

1)	Determine whether the statement 'All even positive integers greater than 2 can be expressed as the sum of two odd prime numbers' is valid using predicate calculus and provide a justification.	**3**	(CO1, L3)
2) a)	Explain how to use a truth table to determine if two propositions are logically equivalent by providing an example with specific propositions.	**3**	(CO2, L2)
	OR		
b)	Discuss how many different ways a polynomial of degree 10 can have its coefficients arranged such that exactly three terms are non-zero. Explain your answer.	**3**	(CO2, L2)
3)	Analyze how the principle of mathematical induction can be applied to prove that the sum of the first 10 natural numbers (1 + 2 + ... + 10) equals 55. Provide a brief explanation and a small calculation to support your answer.	**3**	(CO3, L4)
4)	Determine the number of ways to distribute 5 distinct books among 3 students such that each student receives at least one book. Demonstrate your solution using the pigeonhole principle and rules for counting permutations.	**3**	(CO4, L3)
5) a)	Discuss how to prove that for any real number x, if x is rational, then x^2 is also rational.	**3**	(CO5, L2)
	OR		

b)	Discuss with an example, how to prove that a given function has a specified domain.	**3**	(CO5, L2)
6)	Analyze and explain how many different ways can 10 letters be arranged such that two specific letters are never adjacent. Use logical reasoning to justify your answer.	**3**	(CO1, L4)
7) a)	Determine whether the set of all positive integers less than 100 that can be expressed as a sum of two odd positive integers is countable. Justify your answer.	**3**	(CO2, L3)
	OR		
b)	Determine whether the relation on the set = {1, 2, 3, 4} defined by = {(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4), (4,3)} is symmetric. Justify your answer.	**3**	(CO2, L3)
8)	Construct a proof by strong induction to show that for all positive integers n, the number of subsets of a set with n elements is given by 2^n.	**3**	(CO3, L3)
	Section B		**8*7=56**
9)	A function is defined as , where and are real numbers. (i) Determine the domain of this function. (ii) If we consider the specific case where , find the range of values for such that . (iii) Given that , calculate the number of possible integer pairs (a, b) if the function is to map every element in its domain to a unique real number.	**8**	(CO4, L3)
10) a)	Analyze the following statements about divisors and congruences, and provide a logical argument to support your analysis in each part. (i) Prove that if an integer n is divisible by both 3 and 5, then it must also be divisible by 15. (ii) Investigate whether the statement 'Among any three consecutive integers, at least one of them will have a divisor greater than 2' is always true, sometimes true, or never true. Provide examples to support your analysis.	**8**	(CO5, L4)
	OR		
b)	Analyze and prove using logical arguments: (i) If is an even integer, then its square is also an even integer. (ii) Investigate whether the statement 'If a number is divisible by 4, then it is necessarily even' can be logically derived from the first statement without additional assumptions.	**8**	(CO5, L4)
11)	Discuss the concept of equivalence relations and provide an example where the equivalence classes are finite. Additionally, explain how to use a directed graph to determine if a relation is reflexive and symmetric. Finally, illustrate these concepts using a specific set and relation.	**8**	(CO1, L2)
12) a)	Using propositional logic, solve the following problem: (i) Construct a truth table for the compound proposition $\Sigma = (P \rightarrow Q)$ $(\neg P$ R). (ii) Determine whether the statement 'If P is true and Q is false, then R must be true' can be derived from the given compound proposition. Justify your answer by referring to specific rows in your truth table. (iii) Using your truth table, show that the statement Σ is logically equivalent to $((P$ Q) $(\neg P)) \rightarrow R$. Demonstrate this by proving the biconditional relationship using truth table methods.	**8**	(CO2, L3)
	OR		
b)	Demonstrate that for all real numbers x and y, if $x \leq y$ and y is an integer, then $x \leq y$. Furthermore, prove by contradiction that there does not exist any real number z such that $z > 2z + 1$.	**8**	(CO2, L3)

13)	Analyze how a given symmetric matrix can be decomposed into the sum of two matrices that are not necessarily symmetric, using the concept of strong induction. Specifically, (i) construct a proof by strong induction for the claim; (ii) apply your proof to demonstrate the decomposition for a 3x3 symmetric matrix with all diagonal elements as 1 and non-diagonal elements as -1.	**8**	(CO3, L4)
14) a)	Discuss the pigeonhole principle and use it to solve the following problems. For each problem, state clearly how you apply the pigeonhole principle in your solution. (i) Show that among any 10 positive integers, there must be at least two whose difference is a multiple of 9. (ii) Prove that in any set of five distinct positive integers, there exist at least two numbers whose sum is a multiple of 8. (iii) A group consists of 23 people. Explain why it is necessary to have at least 3 people sharing the same birthday (ignoring leap years), and apply the pigeonhole principle to support your argument.	**8**	(CO4, L2)
	OR		
b)	A library has 10 different books of various genres. (i) How many ways can a student choose 3 books such that at least one book is from the mystery genre? Assume 2 out of the 10 books are mysteries. (ii) In how many ways can the student select 5 books if they must include both a mystery and a science fiction book, given there is only 1 mystery and 1 science fiction book in the library?	**8**	(CO4, L2)
15)	A university mathematics club has 10 members (5 males and 5 females). The club decides to form a team of 4 members for an upcoming competition. (i) In how many ways can this be done if at least one female must be included in the team? (ii) If the team is chosen randomly, what is the probability that exactly two males are selected?	**8**	(CO5, L3)
	Section C		**10*2=20**
16)	Consider the statement: For all real numbers and , if and only if . Using this information, (i) prove that if is an integer, then is also an integer; (ii) show that there exists a unique real number such that for any real number ; (iii) reformulate the following argument using predicate logic: 'For every integer , is even if and only if is odd', and determine its validity.	**10**	(CO1, L3)
17) a)	Analyze and prove, using propositional logic, that for any real numbers and , if then . Provide at least two different methods of proof.	**10**	(CO2, L4)
	OR		
b)	Analyze the properties of the relation R defined by R = {(1, 2), (2, 1), (2, 3), (3, 2), (3, 4)} on the set A = {1, 2, 3, 4}. Determine whether R is reflexive, symmetric, transitive, and an equivalence relation. Justify your answers by applying the rules of propositional logic.	**10**	(CO2, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

AI-assisted draft - teacher review and approval required before use